

- Exercises: Starting points sketch with + write
- QM Exp. Success
- Create + Manipulate QM States
- Teach Atomic Scale

(single)

Qubits As Spheres: Bloch Sphere

1 Qubit Pure State:

2-component Complex Vector of Unit Norm $\leftrightarrow$ point on surface of Sphere

$\mathbb{C}_2$ + 1 constraint (normalization) + 1 constraint (global phase) $\rightarrow$ 2 d.o.f.

$$|2\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

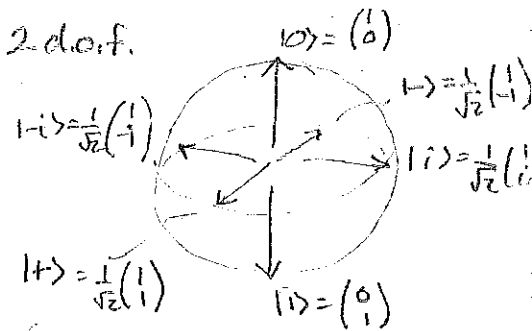
factor of 2 s.t. $|2\rangle(\theta=\pi/2, \phi)$ is on Equator

$$P(0) = \cos^2\frac{\theta}{2} \quad P(1) = \sin^2\frac{\theta}{2}$$

But how to learn ϕ ? (state tomography)

e.g. distinguishing between $|+\rangle$ and $|i\rangle$ and $\frac{|0\rangle + |1\rangle}{\sqrt{2}}$

A: Measure in A second, noncommuting basis.



$|i\rangle$ eigenvectors of Pauli: $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
 $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
 $|i\rangle$ orthogonal $\langle 0|1\rangle = 0 \Rightarrow 180^\circ$ on B.S.

Body-Fixed (Theory): Translate + Read

$$|0\rangle\langle 0| + |1\rangle\langle 1| = |+\rangle\langle +| + |-\rangle\langle -| = \mathbb{I}$$

Identity Insertion for basis change

$$\begin{aligned} |2\rangle &= \mathbb{I}|2\rangle = |+\rangle\langle +|2\rangle + |-\rangle\langle -|2\rangle \\ &= \frac{\cos\frac{\theta}{2} + e^{i\phi}\sin\frac{\theta}{2}}{\sqrt{2}}|+\rangle + \frac{\cos\frac{\theta}{2} - e^{i\phi}\sin\frac{\theta}{2}}{\sqrt{2}}|-\rangle \end{aligned}$$

State has not changed, same vector on B.S.

Space-fixed (Experiment): Always Measure in Z

Map (in software)

$HXH = Z$
Hadamard!

$$\begin{aligned} |+\rangle &\rightarrow |0\rangle & U_{X \rightarrow Z} &= |0\rangle\langle +| + |1\rangle\langle -| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ |-\rangle &\rightarrow |1\rangle \end{aligned}$$

$$H|2\rangle = |2'\rangle$$

$$\langle 0|2'\rangle = \langle +|2\rangle$$

$$\langle 1|2'\rangle = \langle -|2\rangle$$

Pictorially, $\chi_X = [H] \cdot \chi_Z$ + translation
 $\frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & -i \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & i \end{pmatrix}$

For Y measurements, need $U_{Y \rightarrow Z} = |0\rangle\langle i| + |1\rangle\langle -i|$

$$= U_{X \rightarrow Z} U_{Y \rightarrow X} = H(|+\rangle\langle i| + |-\rangle\langle -i|) = H \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} = HS^\dagger$$

$$\text{So, } U_{Y \rightarrow Z} Y U_{Y \rightarrow Z}^\dagger = Z = HS^\dagger Y S H$$

$$HS^\dagger|2\rangle = |2'\rangle, \quad \langle 0|2'\rangle = \langle i|2\rangle, \quad \langle 1|2'\rangle = \langle -i|2\rangle$$

$$\chi_Y = U_{Y \rightarrow Z} \chi_Z = S^\dagger H \chi_Z$$

note reversal of operators in circuit notation

Visualizing Gates:

$Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle = |1\rangle$ global phases do not impact B.S. rep.

$Z|2(\theta, \phi)\rangle = |2(\theta, \phi + \pi)\rangle$ 180° Rotation about Z (polar) axis

$$X|2\rangle = \cos\frac{\theta}{2}|1\rangle + e^{i\phi}\sin\frac{\theta}{2}|0\rangle = e^{i\phi} \left(\cos\left(\frac{\pi-\theta}{2}\right)|0\rangle + e^{-i\phi}\sin\left(\frac{\pi-\theta}{2}\right)|1\rangle \right) = e^{i\phi}|2(\pi-\theta, -\phi)\rangle$$

180° rotation About X axis

$$Y = iXZ \quad \text{s.t.} \quad Y|2\rangle = ie^{i(\phi+\pi)}|2(\pi-\theta, -(\phi+\pi))\rangle$$

Just made the point that global phases are irrelevant in states. Why keeping in Operator Discussion??

1-q. trivial phase $\rightarrow$ logical gate: control promotes global $\rightarrow$ relative phase

recall path-dependent phase in interferometer

e.g. controlled-Y

$$Y = \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} X = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$$

$$\begin{aligned} & \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} \\ & \text{phase kickback} \\ & \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} = S \otimes I \end{aligned}$$

Careful when Building, some resources drop these

Continuous Rotations

$S = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}$: counter clockwise by 90° About Z axis " $\sqrt{Z}$ " $S|4(\theta, \phi)\rangle = |4(\theta, \phi + \pi/2)\rangle$

$$S|+\rangle = |i\rangle \quad ? \quad S = e^{i\pi/4} e^{-iZ\pi/4} = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}$$

$$\begin{aligned} e^{i\pi/4} e^{-iZ\pi/4} |+\rangle &= e^{i\pi/4} \left(\frac{e^{-i\pi/4} |0\rangle + e^{i\pi/4} |1\rangle}{\sqrt{2}} \right) \quad (\text{using Taylor Expansion of exponentiated operator}) \\ &= \frac{e^{i\pi/4} (e^{-i\pi/4} |0\rangle + e^{i\pi/4} |1\rangle)}{\sqrt{2}} \\ &= \frac{|0\rangle + e^{i\pi/2} |1\rangle}{\sqrt{2}} = |i\rangle \end{aligned}$$

$$T = \begin{pmatrix} 1 & e^{i\pi/4} \\ 0 & 1 \end{pmatrix} = e^{i\pi/8} e^{-iZ\pi/8} \quad T|4(\theta, \phi)\rangle = |4(\theta, \phi + \pi/4)\rangle$$

called " $\sqrt{X}$ -gate" ... Historically forgot generator of spin $1/2$ rotations
 $S_j = \sigma_j \frac{1}{2}$ $R_n(\phi) = \exp\left(-\frac{i\vec{S} \cdot \hat{n} \phi}{\hbar}\right)$

Gate Zoo: X, Y, Z, T, S, H, R_x, R_y, R_z: $\{H, T\}$ minimal, sequential produces irrational angles

$\rightarrow$ Error Correction Strategies

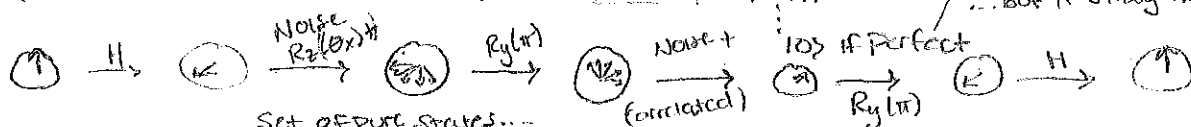
$$= \exp\left(-\frac{i\vec{\sigma} \cdot \hat{n} \phi}{2}\right)$$

Around Effective Axis.

Hahn Spin Echo (~5k citations 6% in past 2yr) 1950 in context of NMR

(Pictorially for now)

$$|0\rangle \xrightarrow{H} \text{[WAIT]} \xrightarrow{H} |0\rangle$$



... but if stray mag. fields...

Set of pure states... will need density matrix to mathematically describe this ensemble.

$$\rho = \sum_i P_i |4_i\rangle\langle 4_i|$$

$$\text{for } |4_i\rangle = e^{-iZ\theta_i}|t\rangle$$

Key:

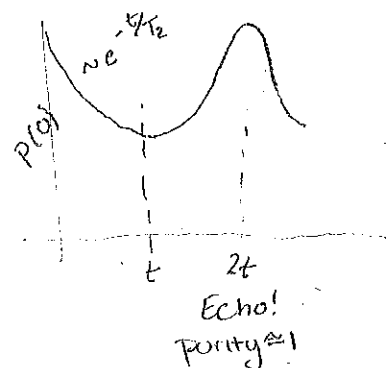
$$R_y(\pi) R_z(\theta) R_y(\pi) = R_z(-\theta)$$

$$\exp\left[-\frac{iY\pi}{2}\right]$$

$$YZY = -Z$$

"Refocussing" / "Dynamical Decoupling"

for coherent errors



Will Gain tools to Analyze in Coming Weeks.

(Error channels + Density Matrices)